

डेबिट शेष	मात्रा (₹)	क्रेडिट बैलेंस	मात्रा (₹)
भूमि एवं भवन लागत मूल्य पर	5,25,000	इक्विटी शेयर पूंजी (प्रत्येक ₹10)	10,00,000
खरीदारी	43,85,000	सामान्य आरक्षित	2,90,000
बकाया भुगतान (20,000 शेयरों पर @ 5 प्रति शेयर)	1,00,000	फर्नीचर पर मूल्यहास का प्रावधान	1,50,000
विक्रय व्यय	11,00,000	व्यापार देयताएँ	2,50,000
अग्रिम कर भुगतान	12,25,000	बिक्री	78,10,000
लागत मूल्य पर पौधा लगाएँ	12,50,000	संयंत्र पर मूल्यहास का प्रावधान	2,50,000
फर्नीचर लागत मूल्य पर	6,50,000	प्रतिभूति प्रीमियम	2,25,000
डिबेंचर पर ब्याज	50,000	लाभ और हानि विवरण में अधिशेष (1-4-2024)	60,000
शुरुआती स्टॉक-इन-ट्रेड	2,25,000	12% डिबेंचर	5,00,000
व्यापार प्राप्तियाँ	4,00,000	अशोध्य ऋणों के लिए प्रावधान	15,000
वेतन	10,00,000	आयोग	50,000
किराया	60,000	कर के लिए प्रावधान	6,25,000
बैंक	2,25,000		
डिबेंचर जारी करने पर छूट	5,000		
अशोध्य ऋण	25,000		
कुल	1,12,25,000	कुल	1,12,25,000

अतिरिक्त जानकारी :

- 31 मार्च, 2025 को समापन स्टॉक ₹10,00,000 था।
- संयंत्र पर 20% प्रति वर्ष और फर्नीचर पर 10% प्रति वर्ष लिखित मूल्य के आधार पर मूल्यहास प्रदान कीजिए।

5. (a) Why are honeybees referred to as eusocial insects? Describe with a relevant diagram how do foraging honeybees communicate among themselves about the food source when a good quality food is located beyond 100 meters away from the hive. 8
- (b) Discuss the infanticide behaviour of males. What tactics are adopted by females to counter the infanticide by the males? 7
6. Write short notes on any **three** of the following : 3×5=15
- Egg rolling behaviour of graylag geese
 - Courtship in three-spined Stickleback Fish
 - Stimulus filtering
 - Inclusive fitness
 - Artificial Intelligence
 - Role of studying animal behaviour in wildlife conservation

[This question paper contains 4 printed pages]

Your Roll No. : 05.06.2026(11)

Sl. No. of Q. Paper : 2044

Unique Paper Code : 2232012403

Name of the Paper : Animal Behaviour -
DSC

Name of the Course : B.Sc. (H) Zoology
(NEP-UGCF)

Semester : IV

Time : 2 Hours Maximum Marks : 60

Instructions for Candidates :

- Write your Roll No. on the top immediately on receipt of this question paper.
 - Attempt **four** questions in all including Question No. 1 which is compulsory.
 - All** questions carry equal marks. **All** parts of a question should be answered together.
1. (a) Define the following (Any **four**) : 1×4=4
- Ethology
 - Kinesis
 - Instinctual behaviour
 - Reasoning

- (v) Intrasexual selection
 (vi) Code breaker
 (b) Differentiate between the following by giving one relevant example each (any **four**) :
 $2 \times 4 = 8$
- (i) Pheromones and Hormones
 (ii) Habituation and Reinforcement
 (iii) Kinesis and Taxis
 (iv) Simple Reflex and the Conditioned Reflex
 (v) Polyandry and polygyny
 (c) Match the List-I with List-II : $0.5 \times 6 = 3$

List-I	List-II
(Scientists)	(Contributions)
(i) Konrad Lorenz	(a) Classical Conditioning
(ii) Ivan Pavlov	(b) Operant Conditioning
(iii) Karl von Frisch	(c) Tail wagging dance in honeybees
(iv) Robert Trivers	(d) Kin Selection
(v) W.D. Hamilton	(e) Imprinting
(vi) B.F. Skinner	(f) Reciprocal Altruism

2. (a) Define communication. Critically analyse the significant roles of tactile, auditory and visual communication by giving two appropriate examples each. 8
 (b) Discuss in detail how is the Fixed action pattern differed from an imprinted or a conditioned response by giving relevant examples. 7
- 3 (a) What is courtship and describe the adaptive significance of the courtship behaviour? Describe briefly the courtship rituals in an insect and a bird. 8
 (b) What is an altruistic behaviour? How is altruistic behaviour explained by kin selection? 7
- 4 (a) What do you understand by the term learnt behaviour? Describe in detail by giving appropriate examples the diversity of learnt behaviours. 8
 (b) What is conditioned reflex? When will you say that conditioning has taken place with respect to a stimulus? Explain by designing an experiment. 7